Numerical Exercise - Regularization

Ridge Regression & Lasso Regression with Gradient Descent

Problem Statement

Fit a linear regression using **Gradient Descent** with learning rate $\alpha = 0.02$ for the data below. The Relative Risk of Coronary Heart Disease (RR-CHD) is believed to depend only on **Diastolic Pressure** (x_1) and **BMI** (x_2).

Initial weights: $w_0 = 5, \quad w_1 = w_2 = -0.03$.
Apply **regularization constant** $\lambda = 5$ and run **2 iterations** of GD for both Ridge and Lasso.

Patient	**Systolic (mm Hg)**	**Diastolic x_1 (mm Hg)**	**BMI x_2**	**RR-CHD y**
1	140	80	35	1.81
2	120	80	25	1.22
3	130	100	30	1.71

Step 1 - Model & Update Equations

Model:

$$h(x) = w_0 + w_1 x_1 + w_2 x_2$$

Ridge - Cost Function:

$$J(w) = \frac{1}{n} \sum_{i=1}^{n} \big(h(x^{(i)}) - y^{(i)}\big)^2 + \lambda(w_1^2 + w_2^2)$$

Ridge - Gradient Update Rules:

$$w_0 \leftarrow w_0 - \frac{\alpha}{n} \sum_{i=1}^{n} \big(h(x^{(i)}) - y^{(i)}\big)$$

$$w_j \leftarrow w_j(1 - \alpha\lambda) - \frac{\alpha}{n} \sum_{i=1}^{n} \big(h(x^{(i)}) - y^{(i)}\big) x_j^{(i)}, \quad j = 1, 2$$

Lasso - Cost Function:

$$J(w) = \frac{1}{n} \sum_{i=1}^{n} \big(h(x^{(i)}) - y^{(i)}\big)^2 + \lambda \sum_{j} |w_j|$$

Lasso - Gradient Update Rules:

$$w_0 \leftarrow w_0 - \frac{\alpha}{n}\sum_{i=1}^{n}\big(h(x^{(i)}) - y^{(i)}\big)$$

$$w_j \leftarrow w_j - \frac{\alpha}{n}\sum_{i=1}^{n}\big(h(x^{(i)}) - y^{(i)}\big)x_j^{(i)} - \alpha\lambda \cdot \mathrm{sign}(w_j), \quad j = 1, 2$$

where $\mathrm{sign}(w_j) = -1$ if $w_j < 0$; 0 if $w_j = 0$; $+1$ if $w_j > 0$.

Step 2 - Initial Predictions & Errors (Both Methods)

Using $w_0 = 5,\ w_1 = -0.03,\ w_2 = -0.03$:

Patient	x_1	x_2	$h(x)$	**Error** $= h(x) - y$
1	80	35	$5 - 0.03(80) - 0.03(35) = 1.55$	-0.26
2	80	25	$5 - 2.40 - 0.75 = 1.85$	$+0.63$
3	100	30	$5 - 3.00 - 0.90 = 1.10$	-0.61

$$\text{Errors} = [-0.26,\ +0.63,\ -0.61]$$

Note: $h_1 = 5 - 0.03 \times 80 - 0.03 \times 35 = 5 - 2.40 - 1.05 = 1.55$*, giving error* $1.55 - 1.81 = -0.26$.

RIDGE REGRESSION

Step 3 (Ridge) - Iteration 1: Update Weights

Intermediate sums:

$$\begin{aligned}
\sum e_i &= -0.26 + 0.63 - 0.61 = -0.24 \\
\sum e_i\, x_{1i} &= (-0.26)(80) + (0.63)(80) + (-0.61)(100) \\
&= -20.8 + 50.4 - 61.0 = -31.4 \\
\sum e_i\, x_{2i} &= (-0.26)(35) + (0.63)(25) + (-0.61)(30) \\
&= -9.10 + 15.75 - 18.30 = -11.65
\end{aligned}$$

Note: $\alpha\lambda = 0.02 \times 5 = 0.1$, so $(1 - \alpha\lambda) = 0.9$.
Update weights:

$$w_0 = 5 - \frac{0.02}{3}(-0.24) = 5 + 0.0016 = \mathbf{5.0016}$$

$$w_1 = -0.03(0.9) - \frac{0.02}{3}(-31.4) = -0.0270 + 0.2093 = \mathbf{0.1823}$$

$$w_2 = -0.03(0.9) - \frac{0.02}{3}(-11.65) = -0.0270 + 0.0777 = \mathbf{0.0507}$$

Step 4 (Ridge) - Iteration 2: Predictions, Errors & Update

Recompute predictions using $w_0 = 5.0016,\ w_1 = 0.1823,\ w_2 = 0.0507$:

$$h_1 = 5.0016 + 0.1823(80) + 0.0507(35) = 5.0016 + 14.584 + 1.7745 = 21.36$$
$$h_2 = 5.0016 + 0.1823(80) + 0.0507(25) = 5.0016 + 14.584 + 1.2675 = 20.85$$
$$h_3 = 5.0016 + 0.1823(100) + 0.0507(30) = 5.0016 + 18.23 + 1.521 = 24.75$$

Patient	$h(x)$	y	**Error** e_i
1	21.36	1.81	19.55
2	20.85	1.22	19.63
3	24.75	1.71	23.04

Intermediate sums:

$$\sum e_i = 19.55 + 19.63 + 23.04 = 62.22$$
$$\begin{aligned}\sum e_i\, x_{1i} &= 19.55(80) + 19.63(80) + 23.04(100)\\ &= 1564.0 + 1570.4 + 2304.0 = 5438.4\\ \sum e_i\, x_{2i} &= 19.55(35) + 19.63(25) + 23.04(30)\\ &= 684.25 + 490.75 + 691.20 = 1866.2\end{aligned}$$

Update weights (using $1 - \alpha\lambda = 0.9$):

$$w_0 = 5.0016 - \frac{0.02}{3}(62.22) = 5.0016 - 0.4148 = \mathbf{4.5868}$$
$$w_1 = 0.1823(0.9) - \frac{0.02}{3}(5438.4) = 0.1641 - 36.256 = \mathbf{-36.09}$$
$$w_2 = 0.0507(0.9) - \frac{0.02}{3}(1866.2) = 0.0456 - 12.441 = \mathbf{-12.40}$$

Final Weights After 2 Iterations (Ridge): $w_0 = 4.5868, \quad w_1 = -36.09, \quad w_2 = -12.40$

LASSO REGRESSION

Step 5 (Lasso) - Iteration 1: Update Weights

Predictions and errors are the same as Step 2: $[-0.26,\ +0.63,\ -0.61]$.

Since $w_1 = w_2 = -0.03 < 0$: $\text{sign}(w_1) = \text{sign}(w_2) = -1$.

Note: $\alpha\lambda = 0.02 \times 5 = 0.1$.

Intermediate sums (same as Ridge Iteration 1):

$$\sum e_i = -0.24, \qquad \sum e_i\, x_{1i} = -31.4, \qquad \sum e_i\, x_{2i} = -11.65$$

Update weights:

$$w_0 = 5 - \frac{0.02}{3}(-0.24) = 5 + 0.0016 = \mathbf{5.0016}$$
$$w_1 = -0.03 - \frac{0.02}{3}(-31.4) - (0.1)(-1) = -0.03 + 0.2093 + 0.1 = \mathbf{0.2793}$$
$$w_2 = -0.03 - \frac{0.02}{3}(-11.65) - (0.1)(-1) = -0.03 + 0.0777 + 0.1 = \mathbf{0.1477}$$

Step 6 (Lasso) - Iteration 2: Predictions, Errors & Update

Recompute predictions using $w_0 = 5.0016,\ w_1 = 0.2793,\ w_2 = 0.1477$:

$$h_1 = 5.0016 + 0.2793(80) + 0.1477(35) = 5.0016 + 22.344 + 5.1695 = 32.52$$
$$h_2 = 5.0016 + 0.2793(80) + 0.1477(25) = 5.0016 + 22.344 + 3.6925 = 31.04$$
$$h_3 = 5.0016 + 0.2793(100) + 0.1477(30) = 5.0016 + 27.930 + 4.4310 = 37.36$$

Patient	$h(x)$	y	**Error** e_i
1	32.52	1.81	30.71
2	31.04	1.22	29.82
3	37.36	1.71	35.65

Intermediate sums:

$$\sum e_i = 30.71 + 29.82 + 35.65 = 96.18$$
$$\begin{aligned}\sum e_i\, x_{1i} &= 30.71(80) + 29.82(80) + 35.65(100)\\ &= 2456.8 + 2385.6 + 3565.0 = 8407.4\\ \sum e_i\, x_{2i} &= 30.71(35) + 29.82(25) + 35.65(30)\\ &= 1074.85 + 745.50 + 1069.50 = 2889.85\end{aligned}$$

Since $w_1 = 0.2793 > 0$ and $w_2 = 0.1477 > 0$: $\text{sign}(w_1) = \text{sign}(w_2) = +1$.
Update weights:

$$w_0 = 5.0016 - \frac{0.02}{3}(96.18) = 5.0016 - 0.6412 = \mathbf{4.3604}$$
$$w_1 = 0.2793 - \frac{0.02}{3}(8407.4) - (0.1)(+1) = 0.2793 - 56.049 - 0.1 = \mathbf{-55.87}$$
$$w_2 = 0.1477 - \frac{0.02}{3}(2889.85) - (0.1)(+1) = 0.1477 - 19.266 - 0.1 = \mathbf{-19.22}$$

Final Weights After 2 Iterations (Lasso): $w_0 = 4.3604, \quad w_1 = -55.87, \quad w_2 = -19.22$

Interpretation & Comparison

Metric	Ridge	Lasso
Final w_0	4.5868	4.3604
Final w_1	-36.09	-55.87
Final w_2	-12.40	-19.22
Regularization term	$\lambda(w_1^2 + w_2^2)$ - L2	$\lambda(\lvert w_1\rvert + \lvert w_2\rvert)$ - L1
Penalty type	Smooth, squared	Absolute value
Behavior	Shrinks weights proportionally	Can force weights to zero
GD update difference	$-\alpha\lambda w_j \cdot$ (via $1 - \alpha\lambda$)	$-\alpha\lambda\,\text{sign}(w_j)$

Key Insight:

- **Ridge** multiplies each weight by $(1 - \alpha\lambda)$ at every step, shrinking weights proportionally. They rarely reach exactly zero.
- **Lasso** subtracts a *fixed* $\alpha\lambda$ in the sign direction of each weight. This constant push can drive weights to exactly zero, performing automatic feature selection.
- Both methods diverge here because $\lambda = 5$ is large relative to the data scale and predictions quickly overshoot the targets. In practice, λ and α are tuned via cross-validation.
- Lasso penalizes *less aggressively* per iteration than Ridge when weights are large (fixed ± 0.1 vs. $\alpha\lambda w_j$), so Lasso's weights diverge more slowly in magnitude.